

P31, P32, P33, IIC usage instructions

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This process is applicable for reading the digital outputs of P31, P32, and P33, and the slave address 0x58 of chip IIC.

Definition of Line:

Red: VCC Black: GND Yellow line: SDA White line: SCL

After the product is powered on, send the IIC communication command.

The specific steps are as follows:

1. Slave address 0x58, after shifting one bit to the left, write 0 to the lowest bit to obtain the slave address 0xB0 for the write command
2. The MCU continuously sends IICstart, 0xB0, 0x01, 0x01, IICstop (i.e. writes the 0x01 value to the 0x01 address register), and starts the ASIC acquisition command.

S	Slave ID[6:0]	0	A	Register Address[7:0]	A	Data[7:0]	N/NA	P
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3. Delay of 10ms

4. Slave address 0x58, after shifting one bit to the left, write 1 at the lowest bit to obtain the slave address 0xB1 for the read command
5. MCU continuously sends IICstart, 0xB0, 0x04, IICstart, 0xB1, Byte1, Byte2, Byte3 (i.e. reads 3 bytes continuously from 0x04 address)

S	Slave ID[6:0]	0	A	Register Address[7:0]	A	P
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S	Slave ID[6:0]	1	A	Data1[7:0]	A	Data2[7:0]	A
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Data2[7:0]	A	Data3[7:0]	A	...	A/ NA	P
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6. RAW_P=Byte1*65536+Byte2*256+Byte3;

7. If(RAW_P>8388608)

Inter P= RAW_P - 16777216;

else

Inter P= RAW_P;

8. FinalP pressure conversion formula: $P = \text{Inter P} / 2^{21} * \text{range} + \text{zero point}$

(current sample range=1600kPa, zero point=0kPa), unit: Kpa

Note: Different ranges require modification of different range values

See attachment for example code